Supplementary Materials for ADMET-Bench: Task-Dependent Performance of GNNs and Pretrained Models in ADMET Prediction

I. Architecture-Selection Experiments

This supplementary document reports the preliminary architecture-selection screen used to choose the graph neural network backbone for the main study. Eight message-passing variants (GraphConv, GCN, GAT, GraphSAGE, GIN, TAG, SGC, and TransformerConv) were evaluated on all six benchmark endpoints under the same scaffold-split pipeline, target transformations, and evaluation metrics used in the main manuscript. For each architecture and endpoint the reported result is the best of five configurations selected on the validation split (searching over hidden dimension, number of layers, and learning rate), giving $8 \times 6 \times 5 = 240$ training runs in total. Because the screen shares the main study's protocol, the values below are directly comparable to the main-text results; they are provided for transparency and reproducibility and, as discussed in the main text, show that no single architecture dominates across endpoints.

TABLE S1
Preliminary classification architecture screen: validation-split AUC-ROC and F1 (the model-selection basis), each architecture at its best of five validation-selected configurations (search over hidden dimension, number of layers, and learning rate). Best validation AUC per task in bold. Test-set performance was not used for architecture selection.

Architecture	Tox21 (NR-AR)		hERG	
	Val AUC	Val F1	Val AUC	Val F1
GraphConv	0.819	0.480	0.918	0.903
GCN	0.821	0.435	0.912	0.915
GAT	0.819	0.500	0.907	0.871
GraphSAGE	0.854	0.421	0.903	0.918
GIN	0.844	0.500	0.884	0.933
TAG	0.821	0.444	0.892	0.897
SGC	0.829	0.500	0.920	0.875
Transformer	0.778	0.480	0.905	0.881

TABLE S2
Regression architecture screen: average rank across the four ADME regression datasets by validation R^2 (lower rank is better), the basis on which the backbone was selected; each architecture at its best of five validation-selected configurations (search over hidden dimension, number of layers, and learning rate). Per-dataset validation R^2 shown; the final column gives mean test R^2 as a post-hoc reference only (not used for selection). GraphConv (deployed backbone) in bold.

Arch.	Avg Rank	Caco-2	Half-Life	Clear. Hep.	Clear. Mic.	Test R^2	Params	Time (s)
GraphSAGE	3.00	0.597	0.434	0.316	0.175	0.096	662k	7.3
Transformer	3.25	0.606	0.335	0.327	0.208	−0.197	316k	16.4
TAG	4.25	0.582	0.407	0.228	0.279	0.226	1059k	12.2
GraphConv	4.25	0.575	0.460	0.267	0.081	−0.673	81k	7.1
GCN	4.75	0.534	0.447	0.247	0.137	−0.673	399k	12.8
GAT	5.00	0.513	0.483	0.233	0.122	0.128	1067k	14.9
SGC	5.50	0.560	0.426	0.194	0.159	0.006	463k	12.6
GIN	6.00	0.625	0.368	0.157	0.058	−1.572	113k	6.2

Ranked by validation performance (the only leak-free selection signal), no architecture dominates: the eight backbones span a narrow average regression rank (3.0–6.0; Table S2) and cluster tightly on the classification endpoints (validation AUC 0.78–0.85 on Tox21 and 0.90–0.92 on hERG; Table S1). Like every backbone, GraphConv is unstable on the weakly-learnable clearance and half-life endpoints, an instability that is a property of the endpoints rather than of any single backbone. The validation ranking is moreover a poor predictor of test performance: the best validation-ranked regression backbone (GraphSAGE) reaches only mean test $R^2 = 0.10$, while several validation-competitive backbones (GraphConv, GCN, GIN) post negative mean test R^2 (down to -1.57). GraphConv is mid-ranked on validation (regression validation rank 4.25 of 8) and is not a test-maximizing choice (its mean test R^2 is among the lowest); it

was adopted as a single, fixed, representative message-passing backbone for the main study, and because architecture selection used validation metrics only and left the test split untouched, the single test evaluation reported in the main benchmark remains unbiased.

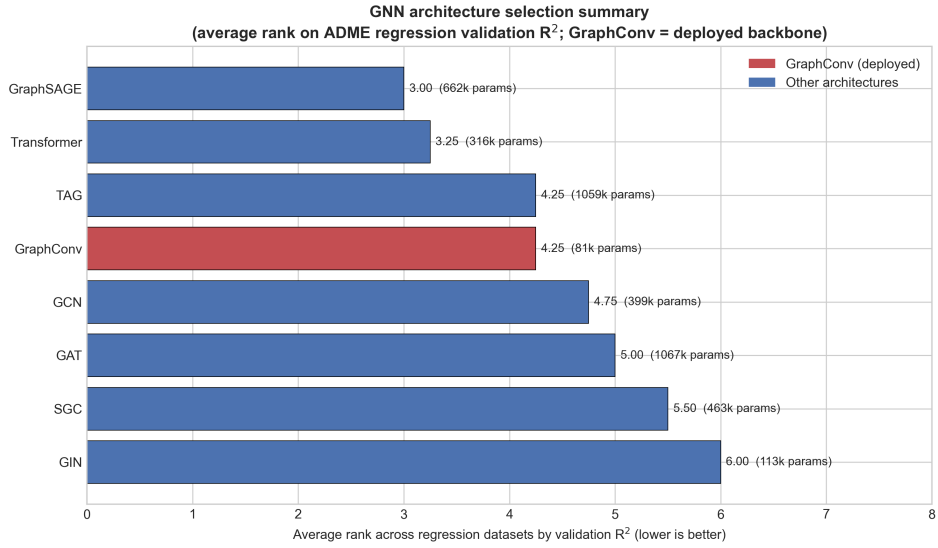


Fig. S1. Preliminary architecture-selection summary showing average rank across the regression datasets by validation R^2 (lower is better); GraphConv (the deployed backbone) is highlighted. No architecture dominates and the per-dataset differences fall within the noise of these small datasets; GraphConv was adopted as a single representative backbone using validation metrics only.

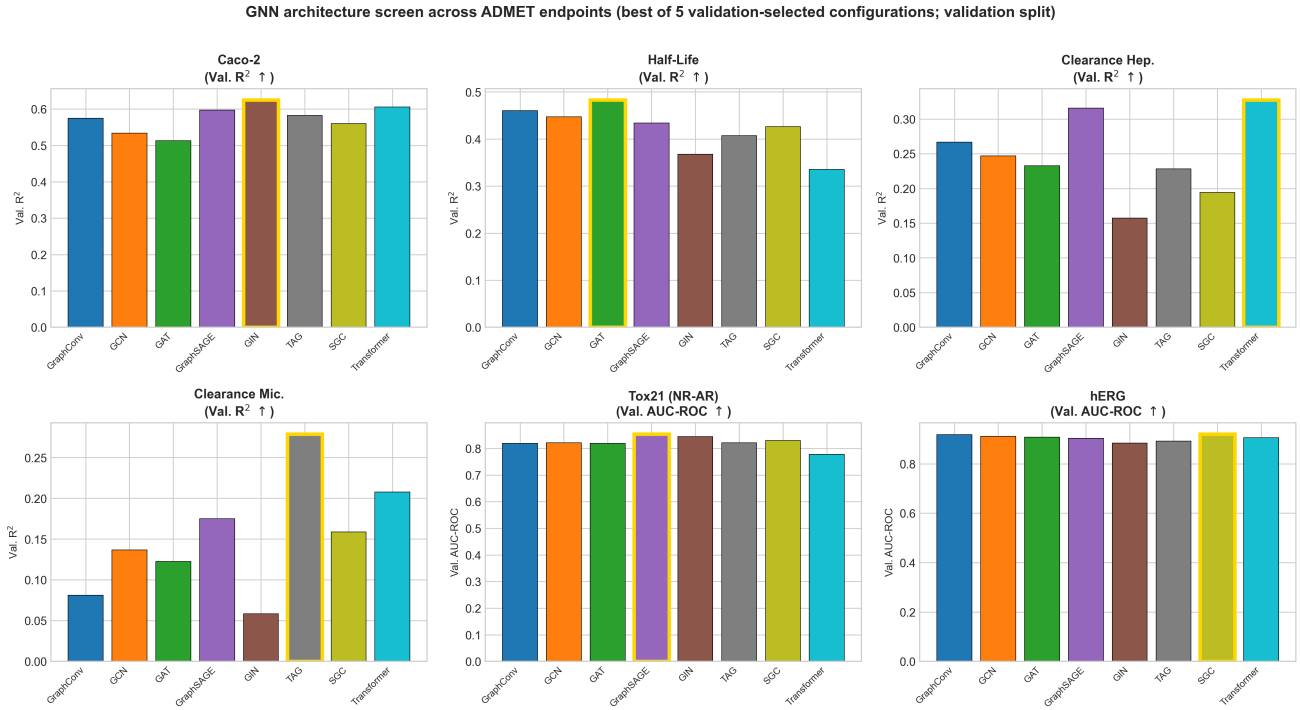


Fig. S2. Per-dataset architecture comparison across all six benchmark endpoints. Classification endpoints (Tox21, hERG) are shown by validation AUC-ROC and regression endpoints (Caco-2, Half-Life, Hepatocyte and Microsomal Clearance) by validation R^2 (the selection basis). The weakly-learnable clearance and half-life endpoints are unstable across all eight architectures, whereas the structure-driven endpoints cluster within a narrow band.